

CT -1
Computer Interfacing (CSE - 405)

Time : 30 minutes

Marks: 20

1. Using a 74ALS138 decoder; interface an 82C55 to the 8088 microprocessor so that it functions at I/O locations: A0H - A6H.
2. Illustrate simple handshaking with appropriate diagrams. Explain how the handshaking process is performed with the printer.

12

$$\begin{array}{r} A6-A0 \\ \hline 3 \\ 08 \\ = 2 \end{array}$$

A0-A1
A2
A4
A6

Military Institute of Science & Technology (MIST)
Department of Computer Science and Engineering
CT - 2, Course: CSE 405: Computer Interfacing

Time: 30 minutes

Total Marks: 20

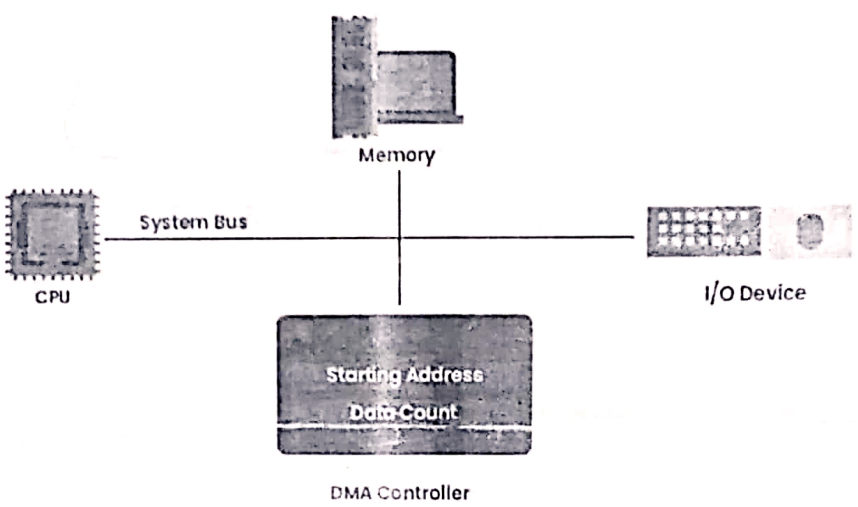
No	Question	Marks																								
1.	"Programmed I/O is not suitable for high-speed data transfer." -use suitable reasoning to explain the following statement.	6																								
2.	Write down the instruction set to transmit a character from the microprocessor via 8251.	4																								
3.	Suppose, <u>3</u> computers are sharing a common SCSI bus among them. <u>3</u> SCSI controllers are connected to the same bus. Now solve the following scenario and find out the value of <u>t</u> at which all the controllers and buses will become free after executing all the tasks given by all the computers. Consider ,data transfer rate between all components = 2.5 sec And arbitration +selection/reselection time= <u>1.8 sec</u> <table border="1"> <thead> <tr> <th>Initiator</th> <th>Target</th> <th>Arbitration time of Initiator (sec)</th> <th>Task time of target given by initiator (sec)</th> </tr> </thead> <tbody> <tr> <td>3</td> <td>3</td> <td>4</td> <td>5</td> </tr> <tr> <td>1</td> <td>3</td> <td>7.2</td> <td>10</td> </tr> <tr> <td>2</td> <td>3</td> <td>11</td> <td>8.4</td> </tr> <tr> <td>2</td> <td>2</td> <td>15</td> <td>12.6</td> </tr> <tr> <td>3</td> <td>2</td> <td>18</td> <td>5</td> </tr> </tbody> </table>	Initiator	Target	Arbitration time of Initiator (sec)	Task time of target given by initiator (sec)	3	3	4	5	1	3	7.2	10	2	3	11	8.4	2	2	15	12.6	3	2	18	5	10
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Military Institute of Science & Technology (MIST)
Department of Computer Science and Engineering
Course: CSE 405: Computer Interfacing
Mid Term Exam

Time: 40 minutes

Total Marks: 30

Part A

No	Question	Marks
1.	In Figure 1, we have a DMA controller. Explain the working principle of the DMA controller and describe in detail which registers are responsible for managing the data count.  Figure 1	8
2.	What are the modes of operation of the 8257 DMA controller?	4
3.	Name three DMA controller data transfer modes and provide detailed information about the Cycle Stealing Mode.	6

Part B

No	Question	Marks
1. ✓	Program the 16550 for operation using <u>8</u> data bits, odd parity, one stop bit and a baud rate of 2400 using 18.432 MHz clock. Assume that the I/O ports are numbered <u>20H</u> and <u>21H</u>. 1 0 0 0 1 0 1 1	9
2.	Explain the difference between maskable and non-maskable interrupt.	3

Military Institute of Science and Technology (MIST)

Department of Computer Science and Engineering

Session: Level 4, Term I (Spring 2024), Batch: CSE 21

Course No: CSE 405

Course Title: Computer Interfacing

CT-03, Date: 16.07.24 (Monday)

Time: 20 minutes

Full Marks: 20

10

1.

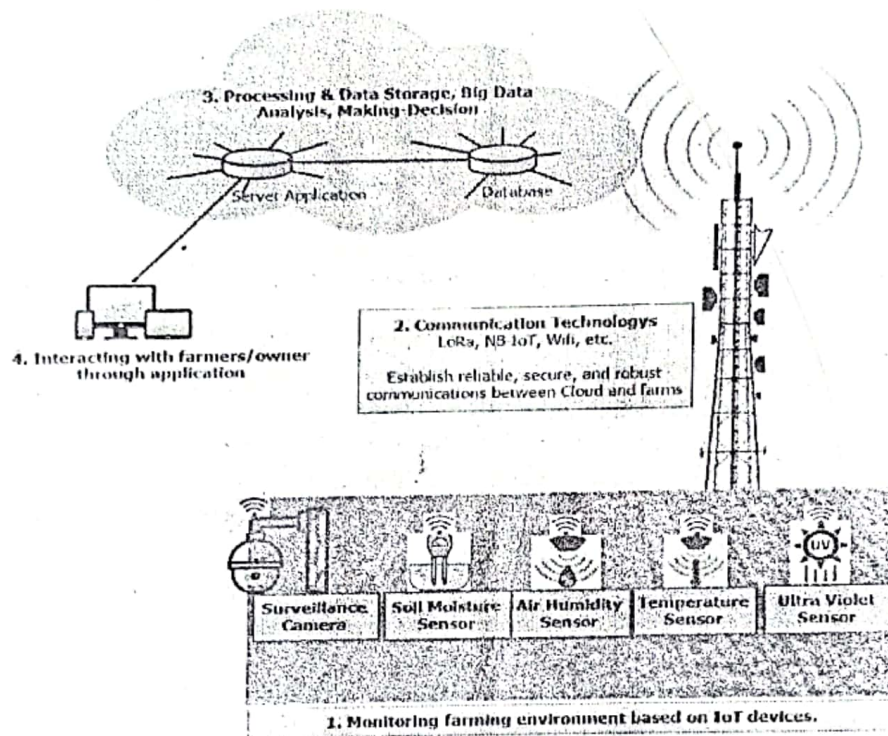


Figure 1 depicts the architecture of IoT ecosystems for smart agriculture. Write down the details of the component and its layer.

2. Define Big Data. Consider an ecommerce system that receives a million requests every day to process payments. There may be several exceptions thrown during these requests such as "payment declined by a payment gateway," "out of inventory," and "invalid address." A developer wants to analyze the last four days' logs to understand which exception is thrown how many times. Illustrate the steps using MapReduce.